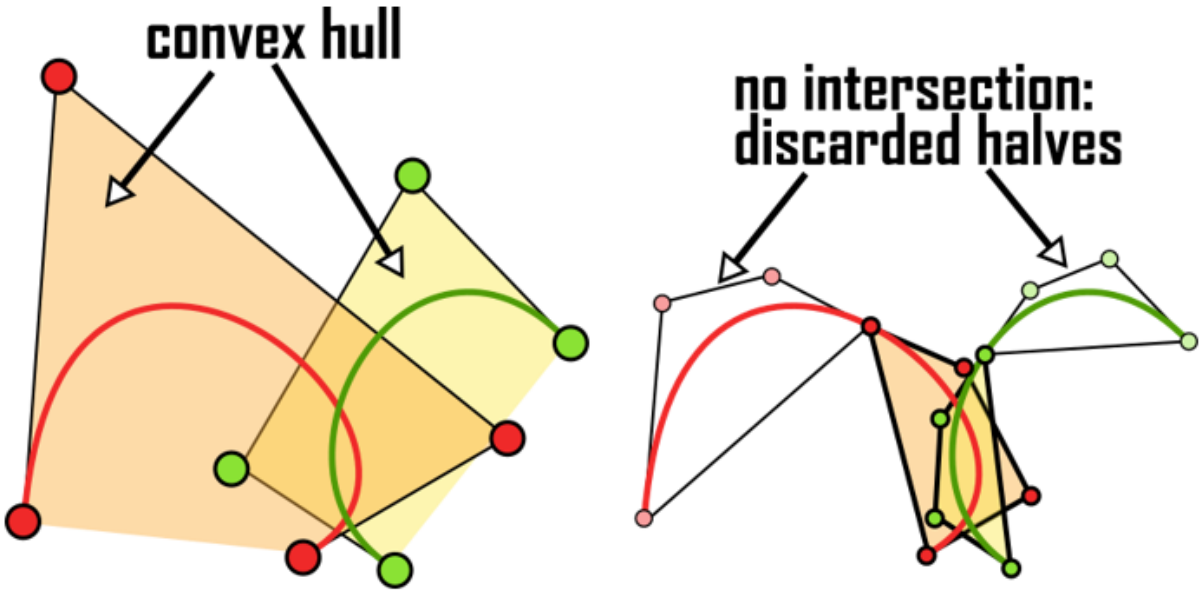


Figure 15.24 The Bézier subdivision method in action (first iteration). The points shown are the actual control points of cubic curves. This method can obviously also be applied to quadratic, quartic or generic curves.

- **Bézier clipping**: for generic Bézier curves, this is the most efficient method, but also the most complicated. We won't get into the details for this one.
- **Interval subdivision** is similar to Bézier clipping, but it adapts better to our stricter requirements. The difference is that, in this case, we first find the vertical and horizontal tangents to the curve. By splitting the curve by making sure that none of the points where its tangent is parallel to one of the axes, we guarantee that in each segment the curve is monotone (because a function can change its trend only in such points) and that for each value point on x, there is only one point belonging to the curve in each one of this sections. Figure 15.25 illustrates how it works for a generic